

towards the other, thereby expediting one-way flow of current.

Over the past decade, researchers have made significant advances towards developing transparent organic solar cells, but they have encountered one persistent stumbling block—finding suitable materials for electrodes that carry current out of the cells. It is rare to find materials in nature that are both electrically-conductive and optically-transparent.

The most widely-used option is indium-tin-oxide (ITO). ITO is conductive and transparent but is also stiff and brittle. So, when the organic solar cell bends, ITO electrodes tend to crack and lift off.

However, indium is expensive and relatively rare. A promising alternative is graphene, a form of carbon that occurs in one-atom-thick sheets and has remarkable characteristics. It is highly-conductive, flexible, robust, transparent and inexpensive. A graphene electrode can be just one nanometre thick—a fraction as thick as an ITO electrode and a far better match for OPV.

Two key problems have slowed the wholesale adoption of graphene electrodes. The first is depositing graphene electrodes onto the solar cell. Deposit-

ing graphene as a top electrode onto the surface of hole transport layer (HTL) is tricky because HTL materials used during deposition process of graphene are soluble in water and sensitive to heat. Any error in the deposition process could destroy HTL materials.

The second problem is that the two electrodes need to play different roles. The ease with which a given material lets go of electrons is a set property called work function. In solar cells, just one of the electrodes should let electrons flow out easily. As a result, having electrodes made of graphene would require changing the work function of one of them so that electrons would know which way to go. But changing the work function of any material is not straightforward.

ITO-free TOPV

Use of a transparent conductive electrode such as ITO limits the reliability and cost of OPV devices, as it is brittle and expensive. Moreover, high sheet resistance of an ITO electrode on flexible substrates limits the maximum width of a single cell. Recent research work has developed an excellent alternative route incorporating ITO-free transparent anode having about 77 per cent transparency based on solu-

tion-processed, highly-conductive PEDOT:PSS in combination with a printed current-collecting grid made of ultra-thin silver metal. When a metal film is only a few nanometres thick, it becomes transparent. Sche-

matic arrangement of an ITO-based/ITO-free device is shown in Fig. 7.

In an ITO-free device, highly conductive polymeric material, which is hundred nanometres thick, like, highly-conductive PEDOT:PSS, is used as a replacement for ITO. However, TOPVs with only PEDOT:PSS electrode do not provide high efficiency for large area devices due to limited conductivity of PEDOT:PSS.

Improving conductivity of such a polymeric electrode is possible by combining it with a honeycomb-shaped silver metal grid patterned by suitably printing in a roll-to-roll processed device (Fig. 8). The printed silver metal grid shows a typical low resistance of one ohm per square. The grid is embedded in a barrier layer—a transparent polymer-based film.

Reactive materials used in OPV are sensitive to water vapour and other chemicals. To avoid deterioration, active material needs to be well-protected from water vapour and other chemicals by using barrier polymer base films. For manufacturing this device, HC PEDOT:PSS is dissolved in an organic solvent and a hundred nanometre-thick film is then deposited on the device by spin coating at 800rpm.

This type of compound anode has a sheet resistance (typically, fifteen-ohm per square) much lower than ITO anode. This is a cost-effective method of producing high-quality devices as the anode is fabricated from solution processing—no litho steps and no high temperature annealing are required.

Metal cathode (LiF/100nm thick Al) is made by thermal evaporation in a vacuum chamber through a shadow mask. It enhances electron injection, reduces interfacial resistance and prevents quenching of excitons. A transparent polythelene-naphthalate (PEN) film (hundred millimetres thick) is used as substrate, as it is strong, good insulator, has better heat transfer properties, and is moisture- and solvent-resistant. Anti-reflection coating is used on the outside surface to reduce reflection, because any light that reflects—potentially as much as ten per cent of the total—does not go through the device. The finished ITO-free device is shown in Fig. 9. **EFY**

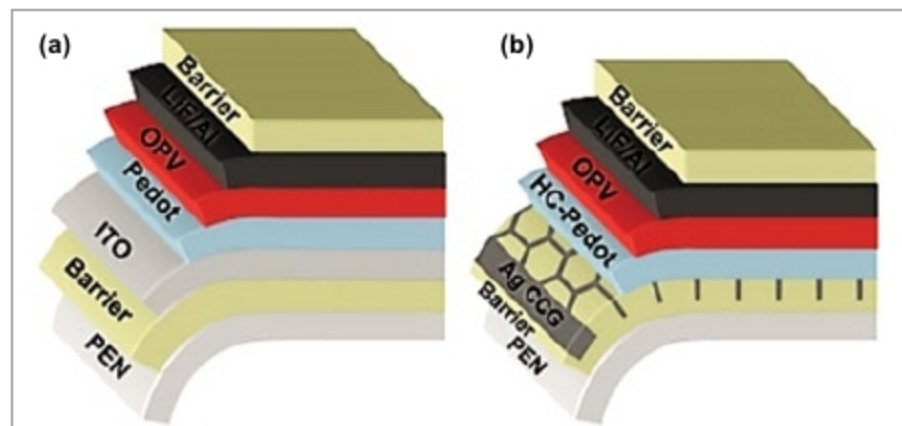


Fig. 7: Schematic arrangement of ITO-based OPV (a) and ITO-free OPV (b)

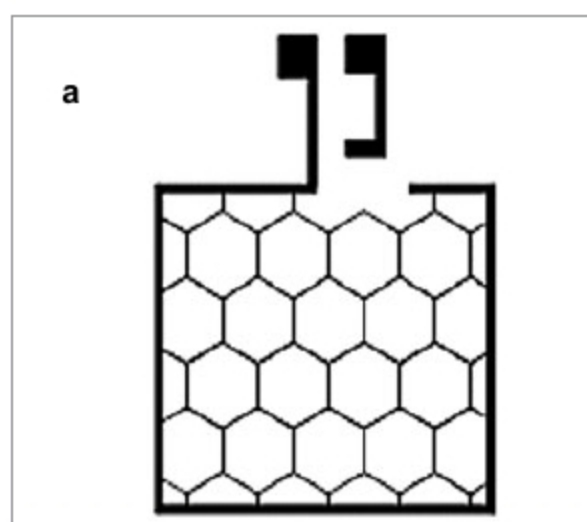


Fig. 8: Conductive grid with honeycomb structure

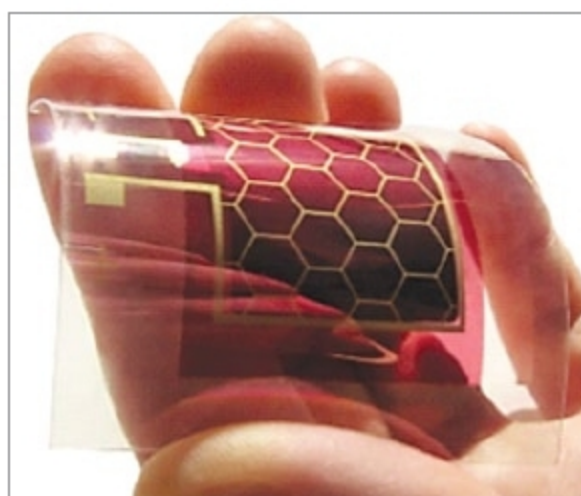


Fig. 9: Finished ITO-free, transparent organic solar cell